Area of Regular Polygons

Find the area of a regular polygon from its triangles, then use it to find a cost.

Name _____ **Date** _____

Tonight's spark

A rooftop garden bed is a regular decagon with 10 sides. It splits into 10 equal triangles. Each has a base of 3 feet and a height of 4 feet. Soil costs $5 per square foot. How much to fill the bed?

Guess first: is it more than $200, or less?

Triangles	Area (square feet)
1	6
10	60

×10

Our class model: one triangle, then multiply

Try it

Work it out on paper. Write one answer on each line.

1. **A regular polygon has 6 sides. One triangle in it has an area of 8. How do you find the polygon's area?**

   A. 8 + 6 **B.** 8 × 6 **C.** 8 ÷ 6 **D.** 8 – 6

   My answer _____

2. **A regular octagon splits into 8 identical triangles of 5 m² each. What is its area?**

   A. 40 m² **B.** 13 m² **C.** 8 m² **D.** 45 m²

   My answer _____

3 **A hexagon patio is 6 triangles. Each triangle has a base of 4 m and a height of 3 m. What is the patio's area?**

A. 12 m² **B.** 72 m² **C.** 6 m² **D.** 36 m²

My answer _____

4 **The patio is 36 m². Tile costs $4 per square meter. How much will the tile cost?**

A. $40 **B.** $9 **C.** $144 **D.** $288

My answer _____

5 **A pentagon design has an area of 60 square feet. Paint costs $2 per square foot. What is the cost?**

A. $120 **B.** $62 **C.** $30 **D.** $240

My answer _____

Talk about it

Why do you multiply one triangle's area by the number of sides instead of adding?

Home mission

Pick a plate or a rug and imagine it as a hexagon of 6 triangles. If one triangle is 5 square units and tile costs $2 per unit, how would you find the cost out loud?

If we get stuck, ask

- How many identical triangles does this shape split into?
- Did you find one triangle, or the whole shape?
- What does the price tell us we multiply by?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-9-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it